When I stop to think that I spent years of my life in chemical laboratories, I have to realize that I ran the risk of accidents. I never worked with really dangerous materials, with tricky explosives, or with deadly poisons—but anything can be dangerous. There could always be explosions, fires, noxious vapors, corrosive chemicals.

Naturally, I had stained my fingers and burned my clothes with acid, and I had cut myself on breaking glassware—but nothing serious ever happened to me in the lab.

Once at Columbia, I remember, a reflux condensation got away from Lloyd Roth while I was standing by. It started bubbling madly and he did the only thing one could do. The first law is GET OUT. He turned off the flame and fled fleetly, and so did I—and then I ran back in to get my books. Nothing happened to me, but if it had, I would have deserved it.

At other times, both in Columbia and in the Navy Yard, there were horror tales of explosions, of hot acid spraying, and so on. It was always on another floor, or when I wasn't there.

One time and one time only did I come close to serious damage— and it was completely my fault. I had a device that determined carbon content in steel powders. It was a quartz tube within which one placed a small alumina boat containing the powdered unknown. The tube was, in turn, placed within a cylindrical electric furnace. A gentle stream of oxygen gas was sent through the tube and into a small U-tube containing a substance that absorbed carbon dioxide. The stream of oxygen combined with the carbon in the heated steel powder to form carbon dioxide, which was added to the absorbing substance as it passed through. By weighing the U-tube before and after, the quantity of car-

bon dioxide, and hence the per cent of carbon in the original sample, could be determined.

On October 5, 1944, a fellow labworker gave me a fluffy white powder and asked me to determine the carbon content. I hadn't the slightest idea as to how much carbon it contained. The thing to do was to start at the safe end—with a very small amount to begin with. If there was then too little carbon to measure, a larger sample could be worked with, and so on.

I did no such thing. With criminal stupidity, I simply filled the boat. Now, if there were a lot of carbon in the sample—as there turned out to be—the combination with oxygen might be rapid enough to explode the substance.